

THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA
FACULTY OF SCIENCE
BACHELOR OF COMPUTER APPLICATIONS (BCA)

BIKE RENTAL EXPLORATORY DATA ANALYSIS

A Project Report on Exploratory Data Analysis Using Python

Submitted By:

GROUP 08

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The Maharaja Sayajirao University of Baroda

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Technologies Used: Python • Pandas • Matplotlib • Visual Studio Code • GitHub

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1. Introduction & Objective

Bike sharing is a modern automated transportation network allowing short-term bicycle rentals. Rental demand fluctuates dynamically based on meteorological conditions, seasonal changes, day of the week, and commute schedules. In this project, the **Bike Sharing Dataset** (UCI Machine Learning Repository) from the Capital Bikeshare system (Washington D.C., 2011–2012) is analyzed using Exploratory Data Analysis (EDA) techniques in Python to uncover operational drivers and usage patterns.

Core Objectives:

1. Understand and describe the structure and attributes of the bike rental dataset.
2. Assess data quality: missing values, duplicates, variable datatypes, and numerical integrity.
3. Compute descriptive statistics including mean, median, standard deviation, and range.
4. Conduct univariate analysis to evaluate individual distribution profiles.
5. Conduct bivariate analysis to examine how rentals vary against temperature, humidity, and windspeed.
6. Compare rental demand across seasons, months, years, weather conditions, and working vs non-working days.
7. Quantify linear relationships through Pearson correlation analysis.
8. Construct informative visualizations (histograms, line charts, bar plots, scatter plots, and heatmaps).
9. Synthesize key empirical findings and operational takeaways.
10. Present clear conclusions for fleet rebalancing and demand forecasting.

2. Dataset Description

2.1 Source: UCI Machine Learning Repository — Bike Sharing Dataset, tracking Capital Bikeshare logs for 2011–2012. The project utilizes two primary data files: **day.csv** (731 daily records, 16 features) and **hour.csv** (17,379 hourly records, 17 features).

2.2 Files Used: day.csv serves as the primary dataset for aggregate demand modeling, seasonal patterns, and weather impacts. hour.csv provides granular hourly validation, intra-day commute surge verification, and data consistency checks.

2.3 Data Structure & Key Variables: The dataset records temporal features (dteday, season, yr, mnth, hr, weekday, holiday, workingday), normalized weather indicators (temp [t/41], atemp [t/50], hum [h/100], windspeed [w/67], weathersit), and count targets (casual, registered, and total rentals cnt).

3. Data Types and Variables

Variable	Data Type	Classification	Description / Value Scale
instant	Integer	Index	Unique record identifier (1 to 731 / 17379)
dteday	Datetime	Date	Observation date (YYYY-MM-DD)
season	Categorical	Discrete	1: Spring, 2: Summer, 3: Fall, 4: Winter
yr / mnth	Categorical	Discrete	Year (0: 2011, 1: 2012) / Month of year (1 to 12)
holiday / workingday	Binary	Indicator Flag	1: Yes / 0: No
weekday	Categorical	Discrete	Day of the week (0: Sun to 6: Sat)
weathersit	Categorical	Ordinal	1: Clear, 2: Mist/Cloudy, 3: Light Rain/Snow, 4: Heavy Rain
temp / atemp	Continuous	Normalized Float	Normalized ambient temp / feeling temp in °C
hum / windspeed	Continuous	Normalized Float	Normalized relative humidity / wind speed
casual / registered	Integer	Numerical Count	Casual leisure rentals vs registered subscriber rentals
cnt	Integer	Target Metric	Total bike rentals (cnt = casual + registered)

4. Tools Used for Analysis

Tool / Library	Version / Environment	Functional Role
Python	3.x	Core computational engine and scripting language.
Pandas	Data manipulation	Data loading, date casting, descriptive statistics, and aggregation.
NumPy	Numerical routines	Matrix calculations, correlation arrays, array transformations.
Matplotlib / Seaborn	Data visualization	Histograms, scatter plots, bar charts, boxplots, heatmaps.
VS Code & GitHub	IDE & Version Control	Code authoring, execution, and project repository sharing.

5. Data Quality Assessment

- **Missing Values:** Evaluated via `isnull().sum()`. Result: **0 missing values** across all 16 variables in `day.csv` and 17 variables in `hour.csv`.
- **Duplicate Records:** Evaluated via `duplicated().sum()`. Result: **0 duplicate rows** identified in both datasets.
- **Data Types:** The `dteday` column was converted from raw string object to ISO datetime using `pd.to_datetime()`.
- **Value Range Validity:** Categorical variables confirmed within designated bounds: `season` ∈ [1, 4], `yr` ∈ [0, 1], `mnth` ∈ [1, 12], `weathersit` ∈ [1, 4].
- **Consistency Verification:** Verified relation `casual + registered == cnt` across all 17,379 hourly records. Result: **0 discrepancies** found.
- **Outlier Screening:** Distribution boxplots revealed high-demand volume spikes on optimal summer days; these represent legitimate peak traffic and were retained.

6. Descriptive Statistics

Parametric and non-parametric statistical summaries were computed over the 731 daily records in `day.csv` using `describe()`.

Metric Variable	Mean	Median	Std Dev	Min	Max
Casual Users	848.18	713.00	686.62	2	3,410
Registered Users	3,656.17	3,662.00	1,560.28	20	6,946
Total Rentals (cnt)	4,504.35	4,548.00	1,937.21	22	8,714
Temperature (norm)	0.495	0.498	0.183	0.059	0.862
Humidity (norm)	0.628	0.627	0.142	0.000	0.973
Windspeed (norm)	0.190	0.181	0.077	0.022	0.507

7. Univariate Analysis

- **Total Rentals (cnt):** Daily count displays a near-symmetric distribution centered at a mean of 4,504 bikes, with a wide standard deviation (1,937 bikes) spanning from a winter low of 22 to a summer peak of 8,714.
- **User Breakdown:** Registered riders represent 81.2% of total volume (averaging 3,656 bikes/day), exhibiting stable commuter behavior. Casual riders average 848 bikes/day with pronounced positive skewness (peaking on weekends and warm holidays).
- **Weather Distributions:** Temperature shows slight bimodality mirroring seasonal transitions. Humidity clusters tightly around 63%, while windspeed is moderately right-skewed.

8. Bivariate Analysis & 9. Grouping and Aggregation

Bivariate and grouped aggregation analyses isolate how environmental and calendar variables influence rental demand:

- **Temperature Association:** Moderate positive correlation ($r = +0.6275$). Higher temperatures encourage cycling, peaking between 20°C and 28°C.
- **Seasonal Pattern:** **Fall (Season 3)** achieves highest daily volume (5,644.30), followed by **Summer (Season 2)** at 4,992.33, **Winter (Season 4)** at 4,728.16, and **Spring (Season 1)** at 2,604.13.
- **Monthly Demand:** Rentals peak in **June (5,772.37)** and **September (5,766.52)**, dropping by 62% in **January (2,176.34)**.
- **Annual Expansion:** Daily rentals grew by **+64.4%** year-over-year from 3,405.76 in 2011 to 5,599.93 in 2012.
- **Weather Conditions:** Clear days (Situation 1) average 4,876.79; mist/cloudy (Situation 2) averages 4,035.86 (-17.2%); precipitation (Situation 3) causes sharp drops to 1,803.29 (-63.0%).
- **Commuter Dynamic:** Working days average 4,584.82 rentals compared to 4,330.17 on non-working days, driven by regular subscriber trips.

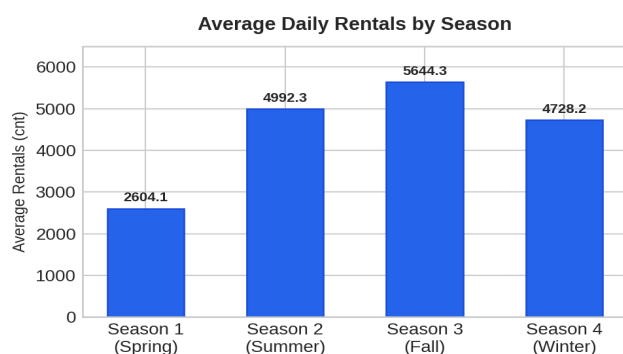


Figure 1: Average Daily Rentals by Season

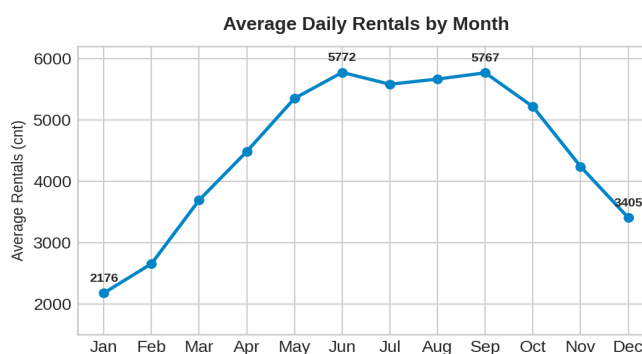


Figure 2: Monthly Rental Demand Curve

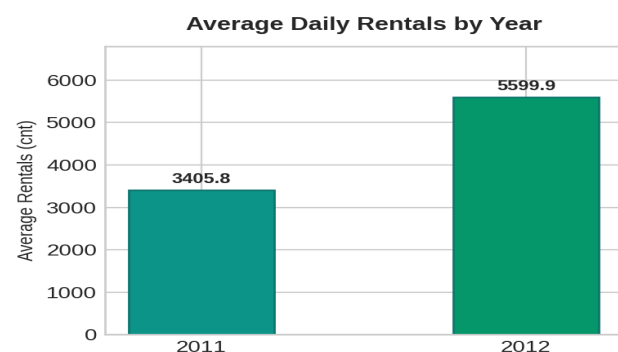


Figure 3: Year-over-Year Growth (2011 vs 2012)

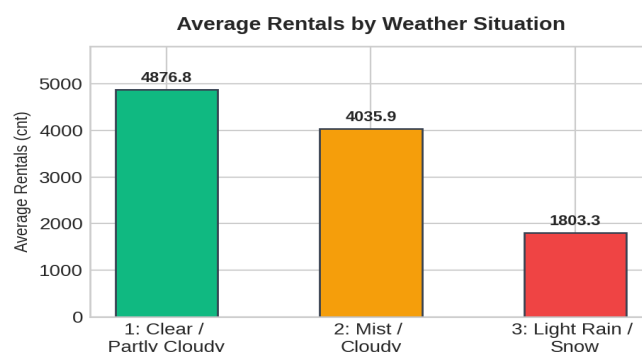


Figure 4: Demand across Weather Situations

10. Correlation Analysis & 11. Visualization

Pearson correlation coefficients (r) quantify linear dependencies among variables. A coefficient r near +1 indicates strong direct proportionality, r near -1 reflects inverse proportionality, and r near 0 indicates linear independence.

Variable Pair	Pearson r	Analytical Interpretation
temp & atemp	+0.9917	Near-perfect collinearity between ambient and feeling temperatures.
registered & cnt	+0.9455	Very strong relationship; subscriber trips dominate overall system load.
casual & cnt	+0.6728	Moderate-to-strong positive correlation; leisure riders scale on sunny days.
temp & cnt	+0.6275	Moderate positive relationship; warmer weather stimulates riding activity.
atemp & cnt	+0.6311	Perceived temperature tracks actual temperature in predicting demand.
windspeed & cnt	-0.2345	Weak negative association; higher winds slightly deter cycling.
hum & cnt	-0.1007	Weak negative association; uncomfortable humidity mildly suppresses trips.

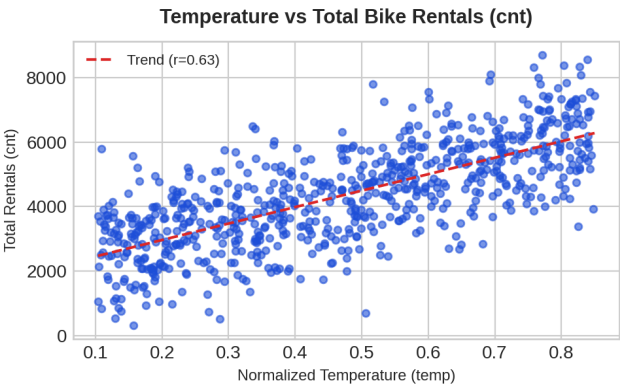


Figure 5: Temperature vs Total Rentals Scatter & Trend

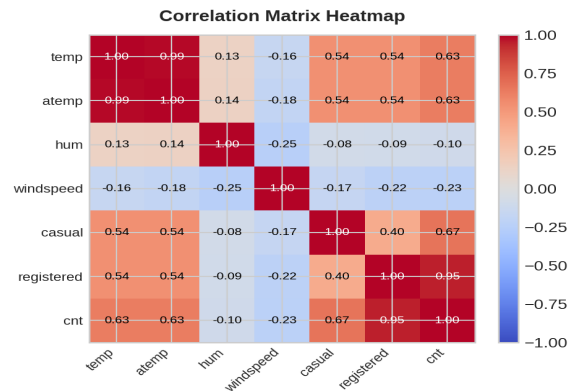


Figure 6: Correlation Matrix Heatmap

Visual Analytics Summary: The visualizations confirm that rental volume responds heavily to temperature and weather clarity. The heatmap clearly isolates multicollinearity between temp and atemp (r=0.99), while showing that registered user demand drives the total count (r=0.95).

12. Key Findings

1. **Flawless Data Integrity:** The dataset contains 731 daily and 17,379 hourly entries with zero missing fields, zero duplicates, and perfect identity consistency (casual + registered = cnt).
2. **Commuter-Driven Baseline:** Registered users account for 81.2% of total ridership (3,656 avg daily rentals), providing a stable core demand.
3. **Substantial Demand Volatility:** Total daily rentals range widely from 22 to 8,714 (mean: 4,504; std: 1,937), demonstrating high sensitivity to external conditions.
4. **Strongest Environmental Predictor:** Temperature displays the strongest positive correlation with rentals ($r = +0.6275$). Comfortable warmth directly stimulates bike usage.
5. **Adverse Weather Deterrence:** Precipitation and poor weather (Situation 3) drop rental volume by 63.0% compared to clear days (1,803 vs 4,877 bikes/day).
6. **Seasonal Hierarchy:** Fall (5,644) and Summer (4,992) generate peak volume, whereas Spring (2,604) experiences a 54% decline due to cold conditions.
7. **Peak Calendar Months:** June (5,772) and September (5,766) record the highest monthly averages, while January records the lowest (2,176).
8. **Rapid Annual Growth:** Daily ridership surged by 64.4% between 2011 (3,406) and 2012 (5,600), reflecting growing bike-sharing adoption.
9. **Mild Atmospheric Drag:** High humidity ($r = -0.10$) and high wind speeds ($r = -0.23$) exert a slight suppressive effect on cycling.
10. **Workday Patterns:** Working days exhibit slightly higher average volume (4,585 vs 4,330) driven by routine morning/evening commuter transit.
11. **Multi-Collinearity Observation:** temp and atemp correlate at 0.9917, indicating redundant information; one can be omitted in predictive modeling.
12. **User Segmentation Differences:** Casual users are highly weather- and weekend-sensitive, while registered users ride consistently across working days.

13. Conclusion

This project successfully conducted an end-to-end Exploratory Data Analysis of the Capital Bikeshare system using Python, Pandas, and Matplotlib. The analysis demonstrates that bike rental demand is strongly governed by seasonality, temperature, and atmospheric conditions, alongside a robust baseline of registered commuters. Data quality verification confirmed zero missing or corrupted values. The insights generated from grouping, bivariate, and correlation analyses provide critical operational guidance for fleet redistribution, dynamic pricing, and maintenance scheduling. Future work can leverage these findings to train predictive machine learning regression models (e.g., Random Forest or Gradient Boosting) for precise demand forecasting.

14. References & 15. Code / GitHub Repository

Academic & Technical References:

1. **UCI Machine Learning Repository:** Fanaee-T, H., & Gama, J. (2013). Bike Sharing Dataset. [UCI Dataset Link](#)
2. **Python Documentation:** Python Software Foundation, Python 3 Standard Library Documentation. [python.org](#)
3. **Pandas Documentation:** McKinney, W. (2010). Data Structures for Statistical Computing in Python. [pandas.pydata.org](#)
4. **Matplotlib Documentation:** Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. [matplotlib.org](#)
5. **Department Guidelines:** Project Details_Div A.pdf, Department of Computer Applications, The Maharaja Sayajirao University of Baroda.

Project Structure & Repository:

Bike Rental EDA Project/

|-- EDA.py

|-- day.csv

|-- hour.csv

|-- Report/

 |-- Bike Rental EDA Report.pdf

GitHub Repository: <https://github.com/Karan9783/bike-rental-eda>

*** End of Project Report ***